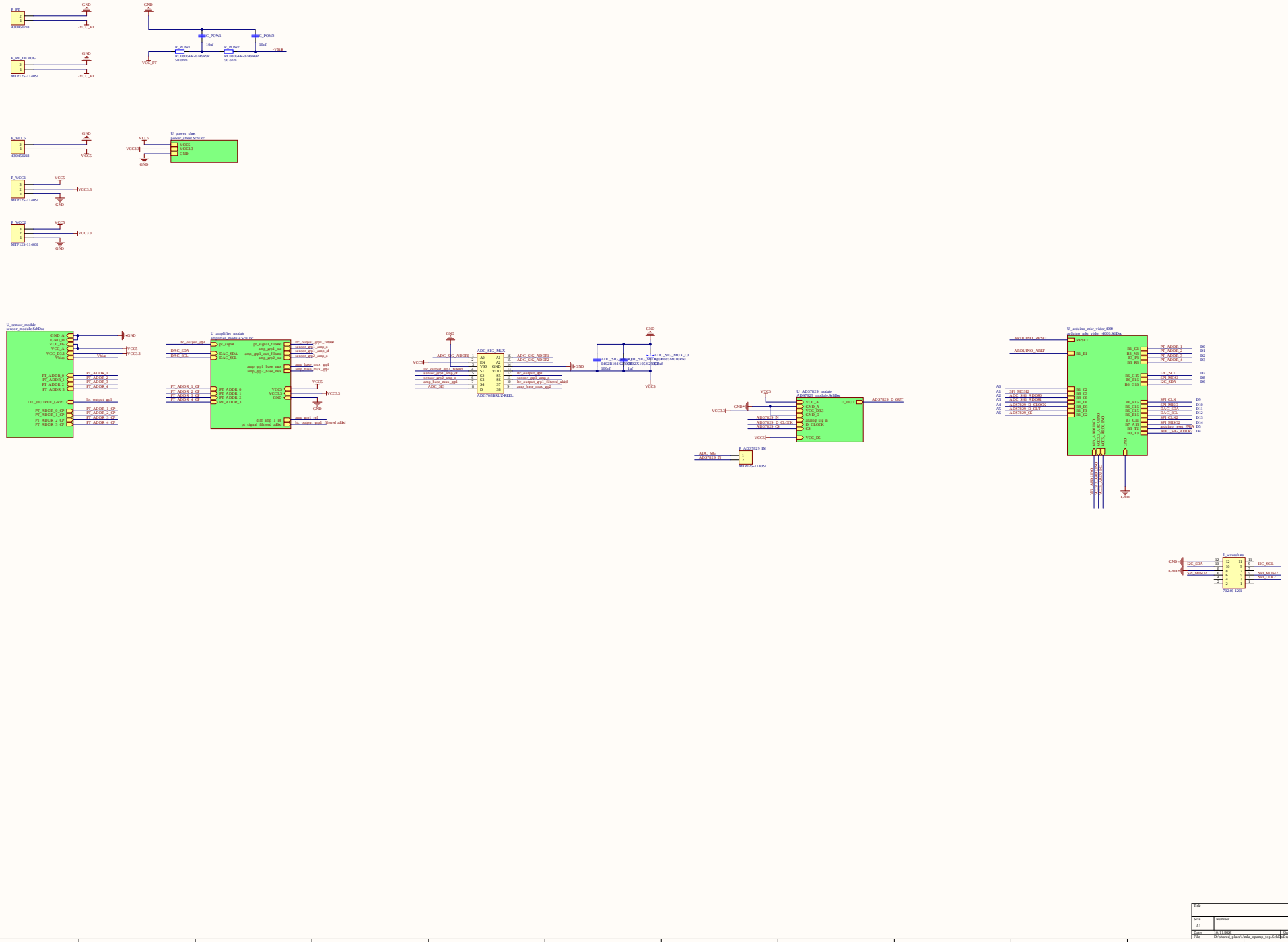
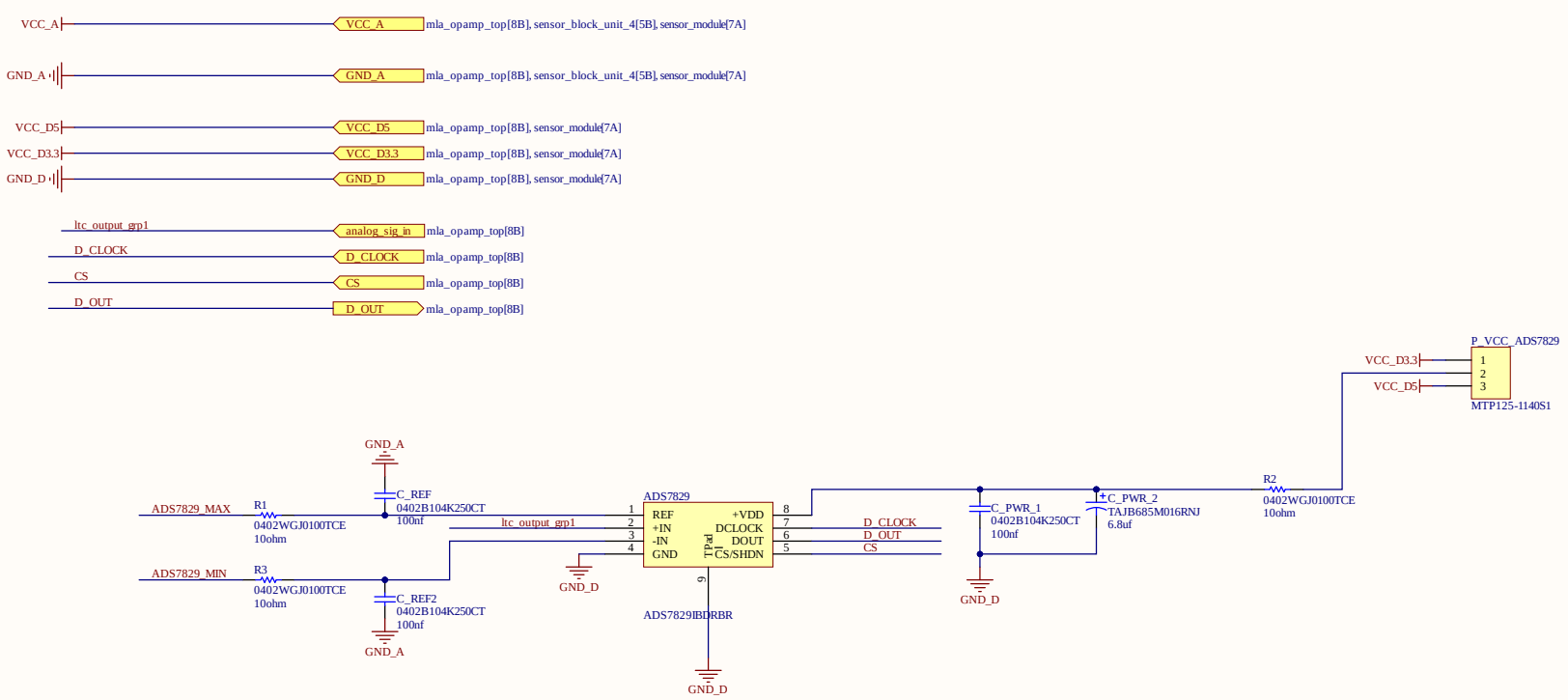
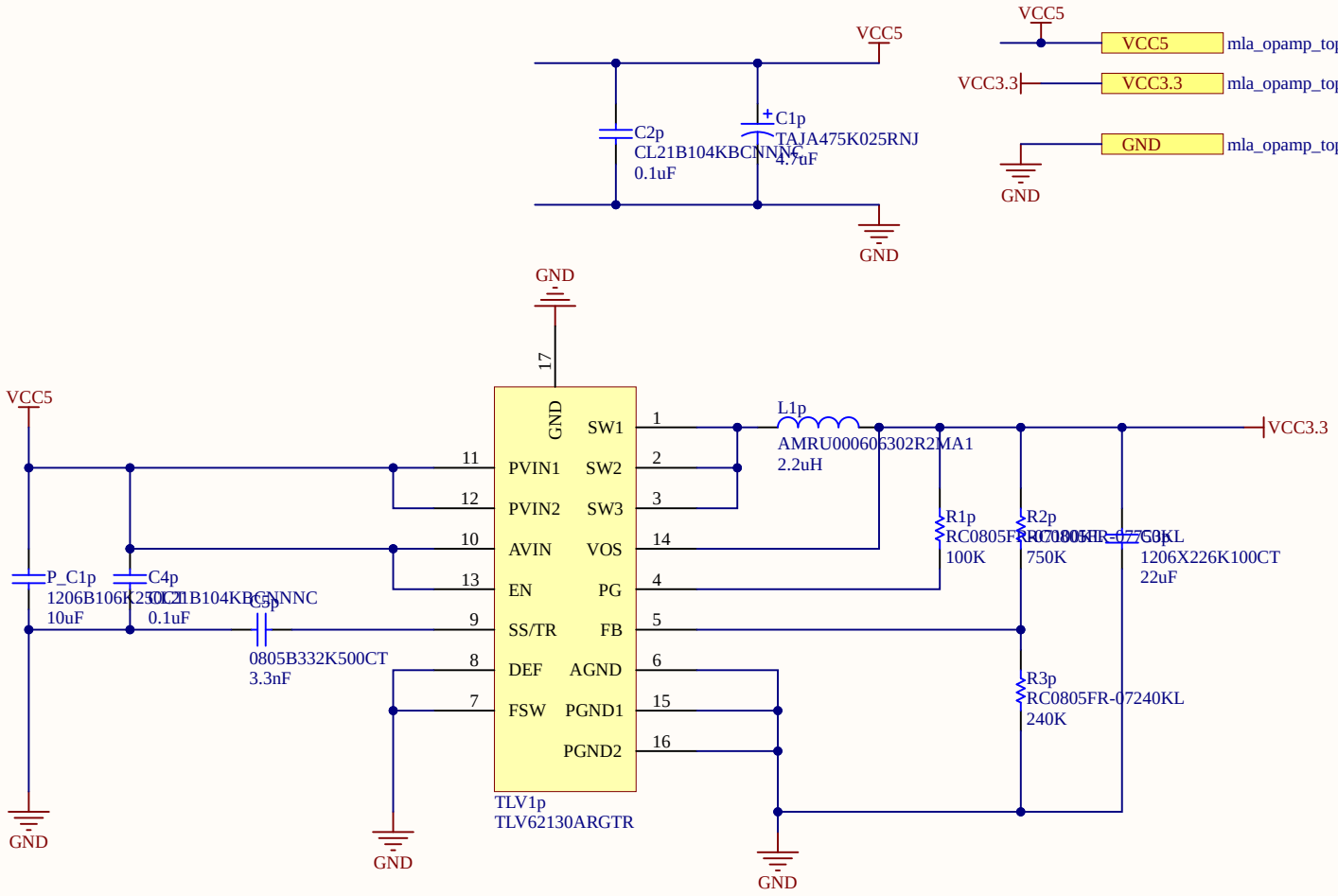
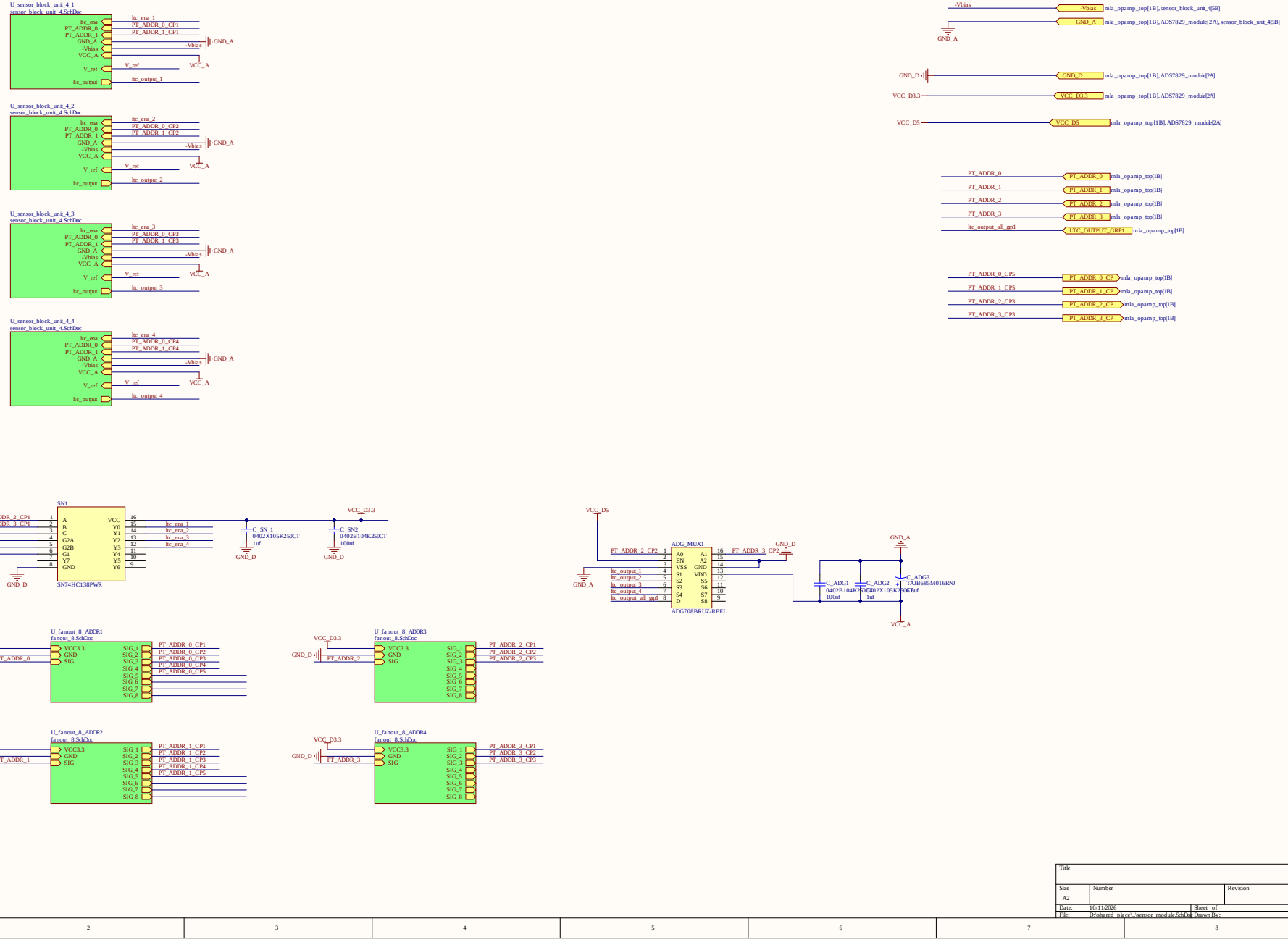

P_PT
P_PT_DEBUG
P_VCC2
P_VCC3
GND
-VCC_PT
C_POW1
C_POW2
10uf
R_POW1
R_POW2
50 ohm
-Vbias
U_power_sheet
power_sheet.SchDoc
VCC2
VCC3.3
U_sensor_module
sensor_module.SchDoc
U_amplifier_module
amplifier_module.SchDoc
ADC_SIG_MUX
ADS7829_module
ADS7829_module.SchDoc
U_arduino_mkr_vidor_4000
arduino_mkr_vidor_4000.SchDoc
ARDUINO_RESET
ARDUINO_AREF
ADC_SIG_ADDR0
ADC_SIG_ADDR1
ADC_SIG_MUX_C1
ADC_SIG_MUX_C2
ADC_SIG
ADS7829_IN
ADS7829_D_CLOCK
ADS7829_CS
ADS7829_D_OUT
P_ADS7829_IN
I2C_SCL
I2C_SDA
SPI_MOSI
SPI_MISO
SPI_CLK
DAC_SCL
DAC_SDA
J_waveshare
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Size
A1
Number
Revision
Date
10/11/2021
Sheet of
File
D:\shared_place_mks_squeap_top.SchDoc
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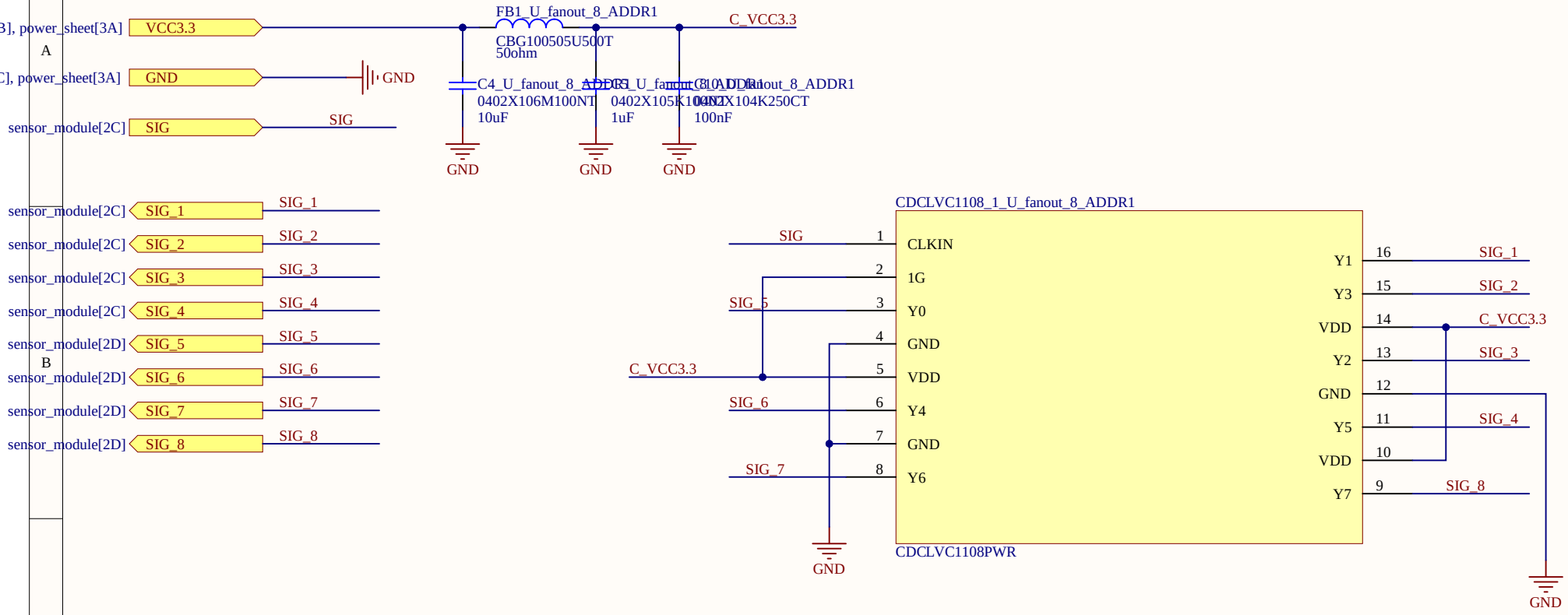


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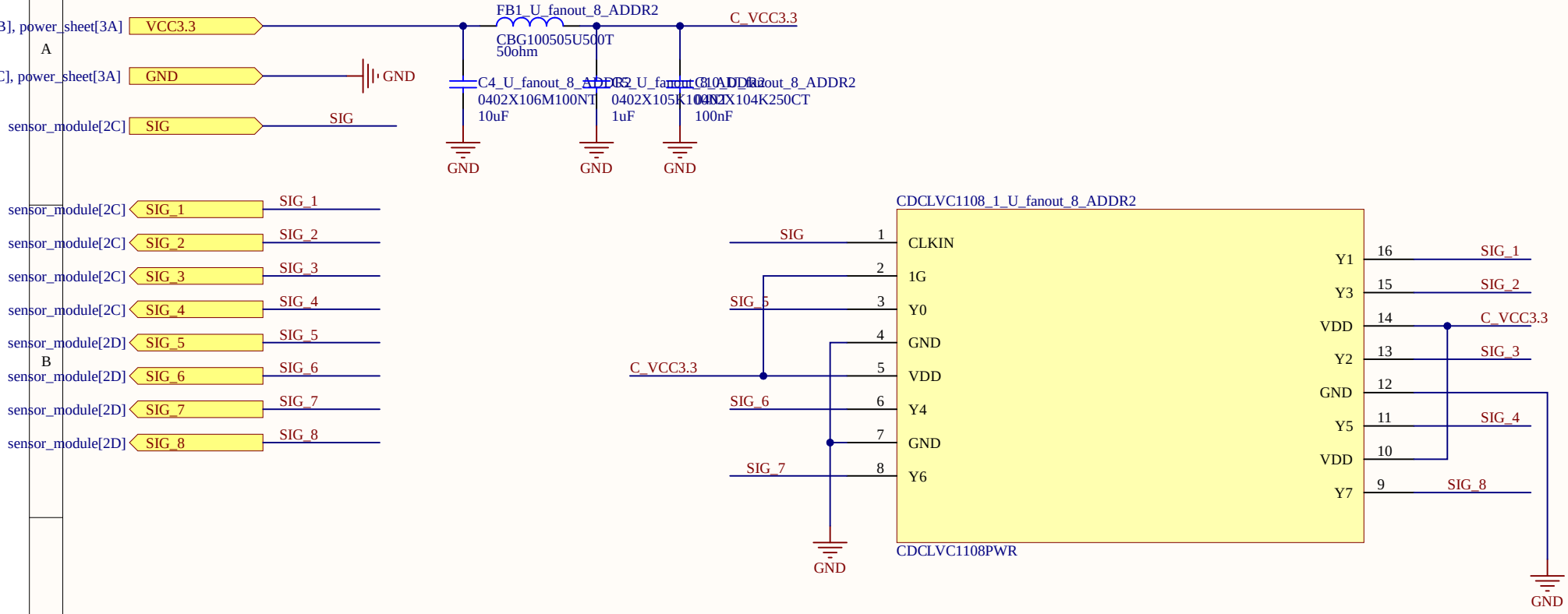


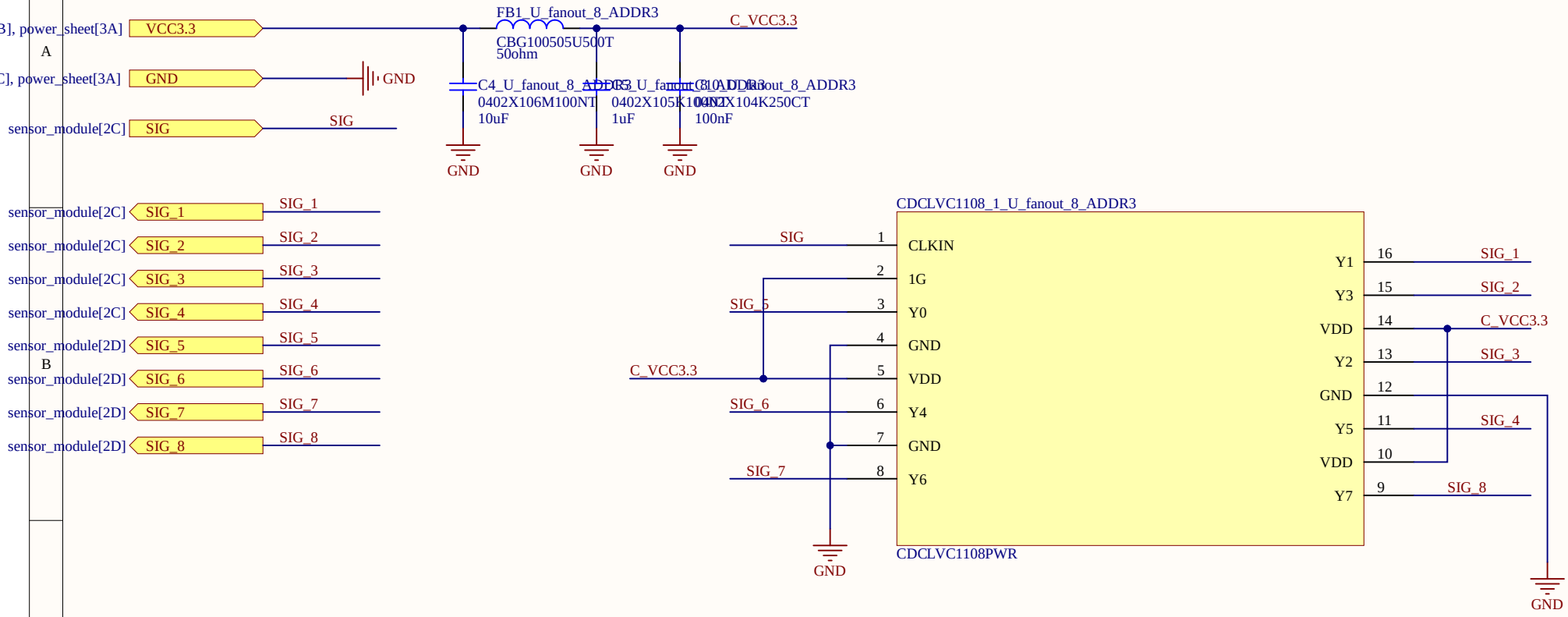
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Size A	Number	Revision
Date: 10/11/2026		Sheet of
File: D:\shared_place\..\power_sheet.SchDoc		Drawn By:


U_sensor_block_unit_4_1
sensor_block_unit_4.SchDoc
ltc_ena
PT_ADDR_0
PT_ADDR_1
GND_A
-Vbias
VCC_A
V_ref
ltc_output
ltc_ena_1
PT_ADDR_0_CP1
PT_ADDR_1_CP1
ltc_output_1
U_sensor_block_unit_4_2
sensor_block_unit_4.SchDoc
ltc_ena_2
PT_ADDR_0_CP2
PT_ADDR_1_CP2
ltc_output_2
U_sensor_block_unit_4_3
sensor_block_unit_4.SchDoc
ltc_ena_3
PT_ADDR_0_CP3
PT_ADDR_1_CP3
ltc_output_3
U_sensor_block_unit_4_4
sensor_block_unit_4.SchDoc
ltc_ena_4
PT_ADDR_0_CP4
PT_ADDR_1_CP4
ltc_output_4
VCC_A
-Vbias
GND_A
GND_D
VCC_D3.3
VCC_D5
PT_ADDR_0
PT_ADDR_1
PT_ADDR_2
PT_ADDR_3
ltc_output_all_grp1
LTC_OUTPUT_GRP1
PT_ADDR_0_CP5
PT_ADDR_1_CP5
PT_ADDR_2_CP3
PT_ADDR_3_CP3
mla_opamp_top[1B]
mla_opamp_top[1B],ADS7829_module[2A],sensor_block_unit_4[5B]
mla_opamp_top[1B],sensor_block_unit_4[5B]
mla_opamp_top[1B],ADS7829_module[2A]
SN1
SN74HC138PWR
C_SN_1
0402X105K250CT
1uf
C_SN2
0402B104K250CT
100nf
ADG_MUX1
ADG708BRUZ-REEL
PT_ADDR_2_CP2
PT_ADDR_3_CP2
ltc_output_1
ltc_output_2
ltc_output_3
ltc_output_4
ltc_output_all_grp1
C_ADG1
0402B104K250CT
100nf
C_ADG2
0402X105K250CT
1uf
C_ADG3
TAJB685M016RNJ
6.8uf
U_fanout_8_ADDR1
fanout_8.SchDoc
U_fanout_8_ADDR2
fanout_8.SchDoc
U_fanout_8_ADDR3
fanout_8.SchDoc
U_fanout_8_ADDR4
fanout_8.SchDoc
VCC3.3
GND
SIG
SIG_1
SIG_2
SIG_3
SIG_4
SIG_5
SIG_6
SIG_7
SIG_8
PT_ADDR_0_CP1
PT_ADDR_0_CP2
PT_ADDR_0_CP3
PT_ADDR_0_CP4
PT_ADDR_0_CP5
PT_ADDR_1_CP1
PT_ADDR_1_CP2
PT_ADDR_1_CP3
PT_ADDR_1_CP4
PT_ADDR_1_CP5
PT_ADDR_2_CP1
PT_ADDR_2_CP2
PT_ADDR_2_CP3
PT_ADDR_3_CP1
PT_ADDR_3_CP2
PT_ADDR_3_CP3
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Number
Revision
Date:
10/11/2026
Sheet of
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Drawn By:

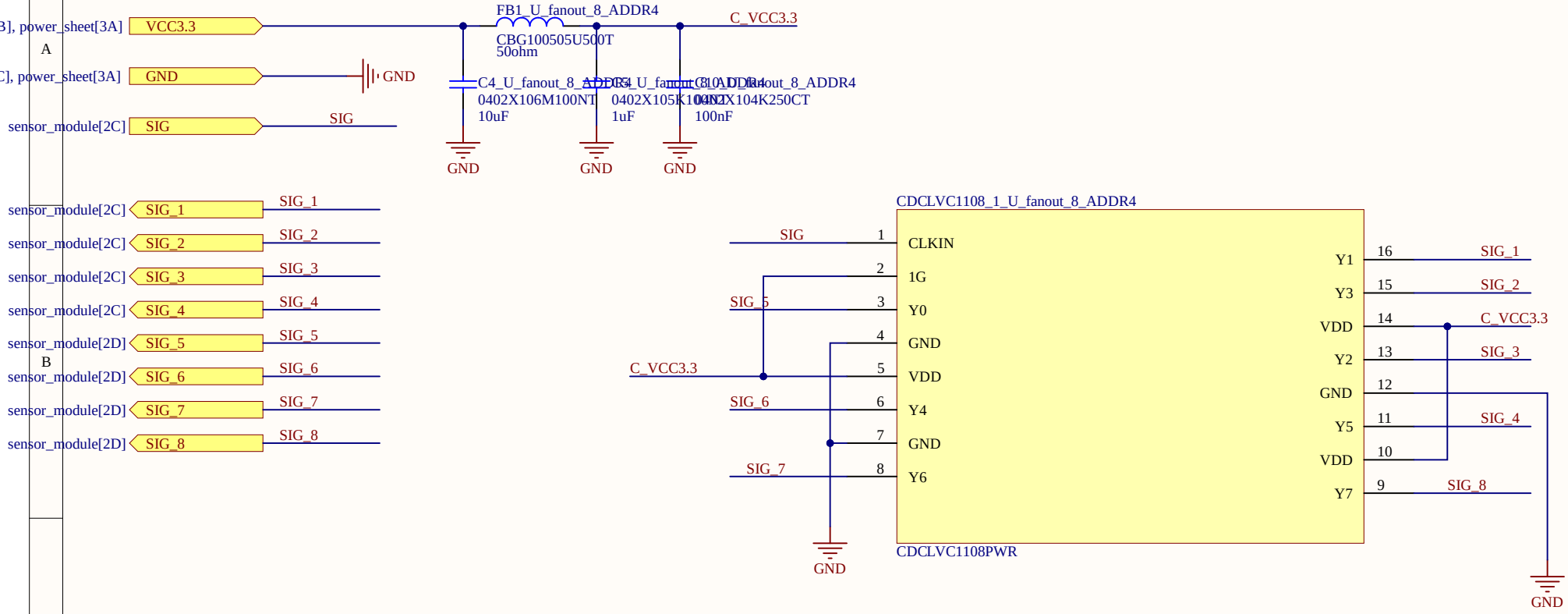


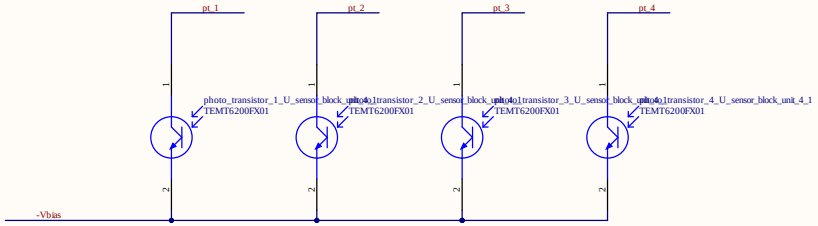
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Date: 10/11/2026		Sheet of
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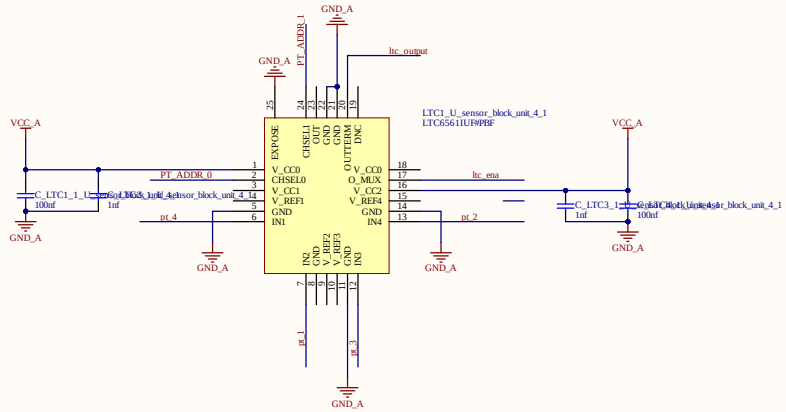

B], power_sheet[3A]
VCC3.3
A
C], power_sheet[3A]
GND
GND
sensor_module[2C]
SIG
SIG
FB1_U_fanout_8_ADDR2
CBG100505U500T
50ohm
C_VCC3.3
C4_U_fanout_8_ADDR2
0402X106M100NT
10uF
C3_U_fanout_8_ADDR2
0402X105K100NT
1uF
C10_U_fanout_8_ADDR2
0402X104K250CT
100nF
GND
GND
GND
sensor_module[2C]
SIG_1
SIG_1
sensor_module[2C]
SIG_2
SIG_2
sensor_module[2C]
SIG_3
SIG_3
sensor_module[2C]
SIG_4
SIG_4
sensor_module[2D]
SIG_5
SIG_5
B
sensor_module[2D]
SIG_6
SIG_6
sensor_module[2D]
SIG_7
SIG_7
sensor_module[2D]
SIG_8
SIG_8
CDCLVC1108_1_U_fanout_8_ADDR2
SIG
1
CLKIN
2
1G
SIG_5
3
Y0
4
GND
C_VCC3.3
5
VDD
SIG_6
6
Y4
7
GND
SIG_7
8
Y6
CDCLVC1108PWR
GND
Y1
16
SIG_1
Y3
15
SIG_2
VDD
14
C_VCC3.3
Y2
13
SIG_3
GND
12
Y5
11
SIG_4
VDD
10
Y7
9
SIG_8
GND
Title
Size
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Number
Revision
Date:
10/11/2026
Sheet of
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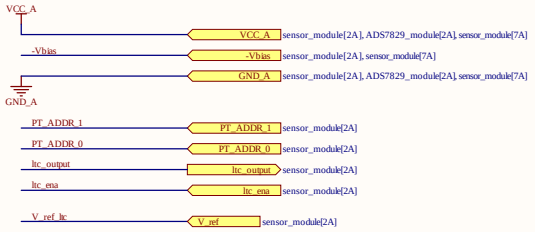


Title		
Size A	Number	Revision
Date: 10/11/2026		Sheet of
File: D:\shared_place\..\fanout_8.SchDoc		Drawn By:


B], power_sheet[3A]
VCC3.3
A
C], power_sheet[3A]
GND
GND
sensor_module[2C]
SIG
SIG
FB1_U_fanout_8_ADDR4
CBG100505U500T
50ohm
C_VCC3.3
C4_U_fanout_8_ADDR4
0402X106M100NT
10uF
C3_U_fanout_8_ADDR4
0402X105K100NT
1uF
C10_U_fanout_8_ADDR4
0402X104K250CT
100nF
GND
GND
GND
sensor_module[2C]
SIG_1
SIG_1
sensor_module[2C]
SIG_2
SIG_2
sensor_module[2C]
SIG_3
SIG_3
sensor_module[2C]
SIG_4
SIG_4
sensor_module[2D]
SIG_5
SIG_5
B
sensor_module[2D]
SIG_6
SIG_6
sensor_module[2D]
SIG_7
SIG_7
sensor_module[2D]
SIG_8
SIG_8
CDCLVC1108_1_U_fanout_8_ADDR4
SIG
1
CLKIN
2
1G
SIG_5
3
Y0
4
GND
C_VCC3.3
5
VDD
SIG_6
6
Y4
7
GND
SIG_7
8
Y6
CDCLVC1108PWR
GND
Y1
16
SIG_1
Y3
15
SIG_2
VDD
14
C_VCC3.3
Y2
13
SIG_3
GND
12
Y5
11
SIG_4
VDD
10
Y7
9
SIG_8
GND
C
D
Title
Size
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Number
Revision
Date:
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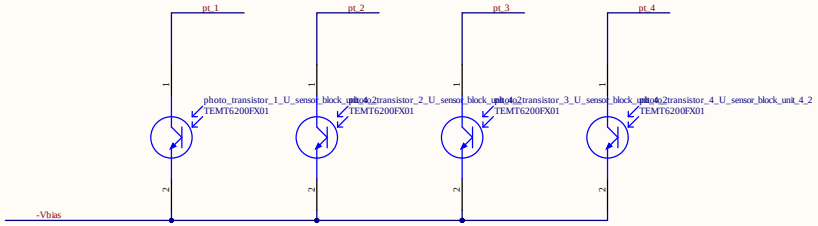


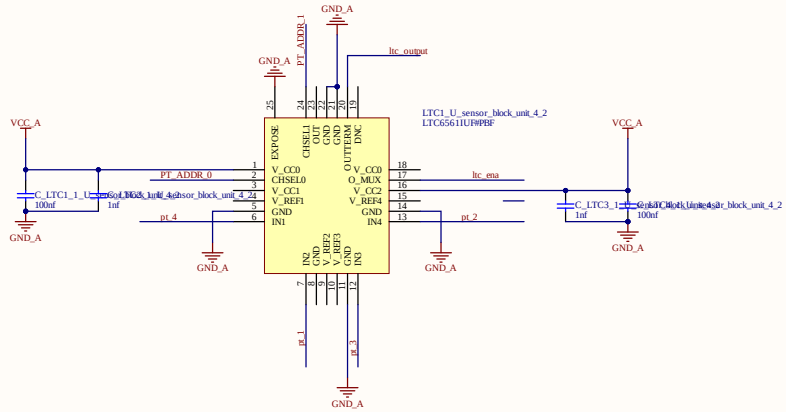


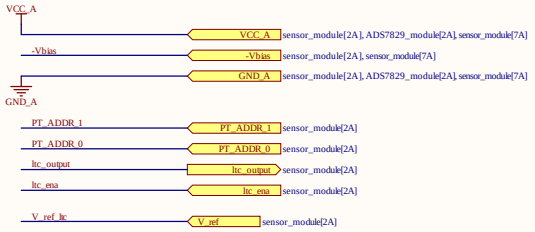


after selection, wait 30ns

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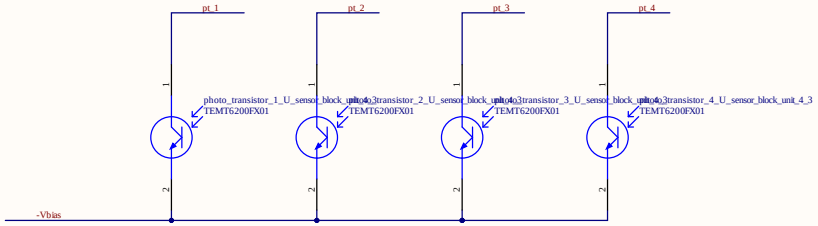


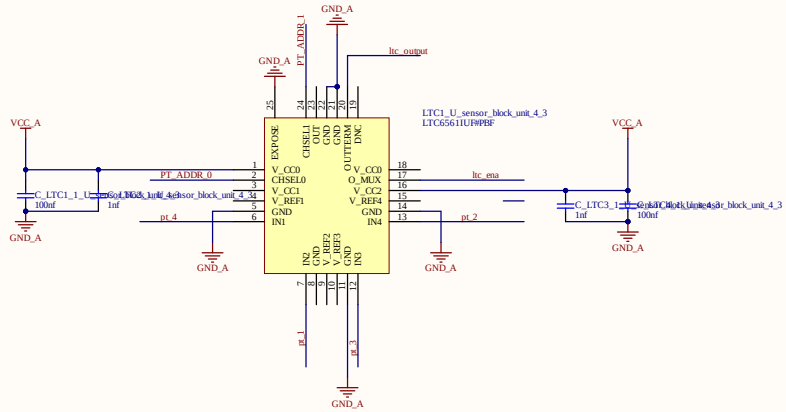


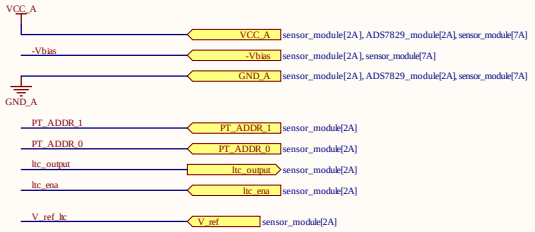


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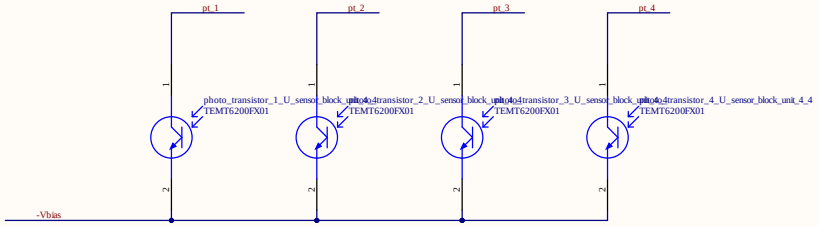


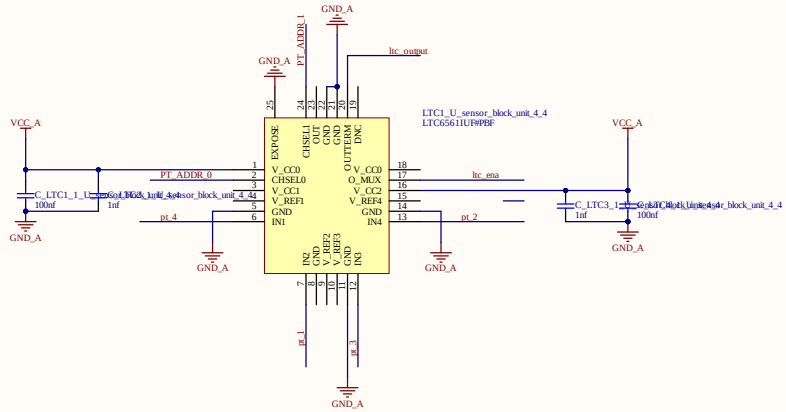


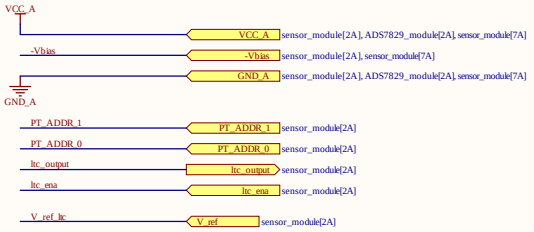


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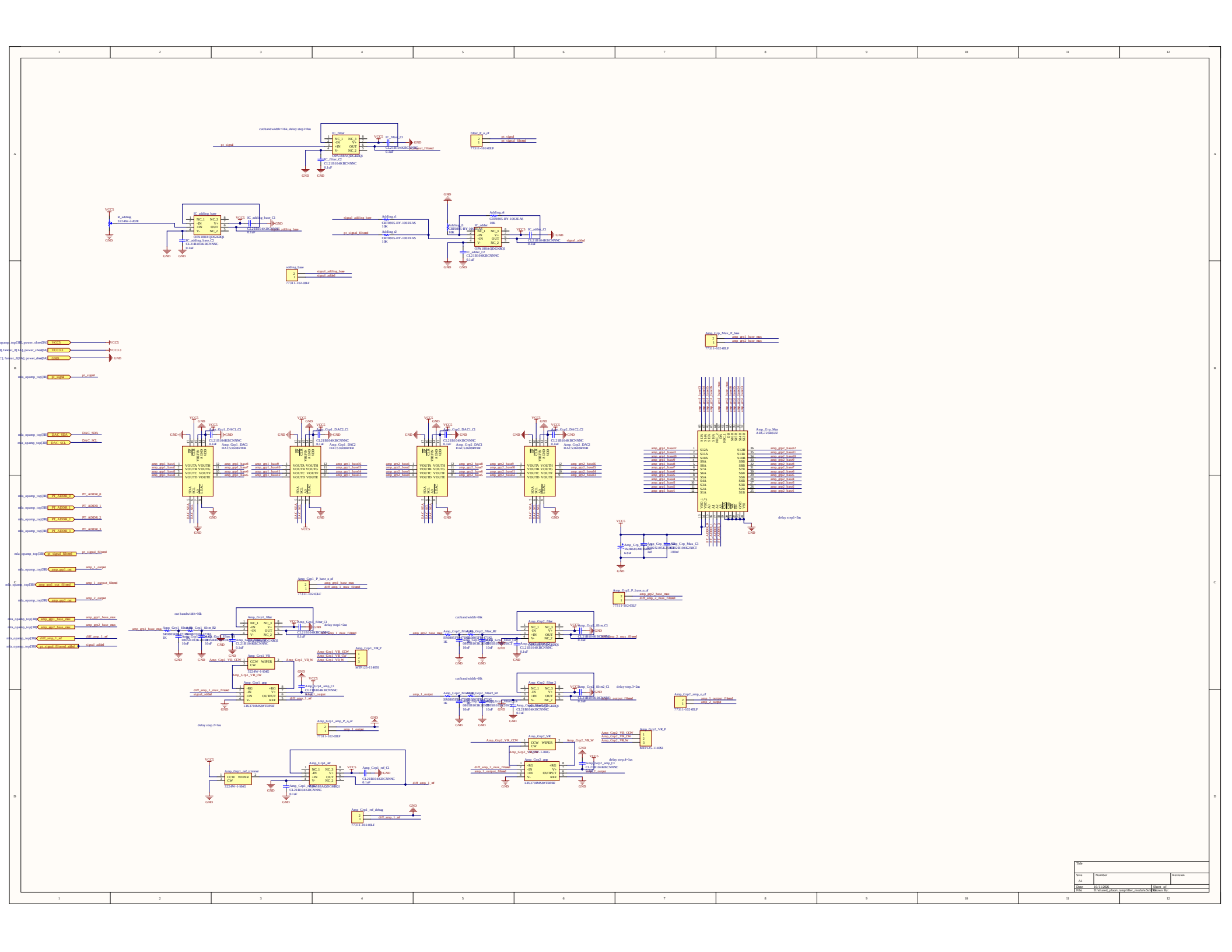


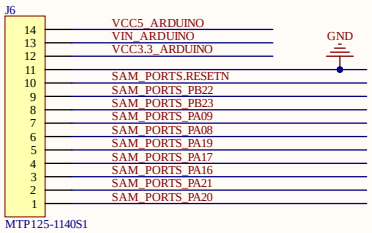


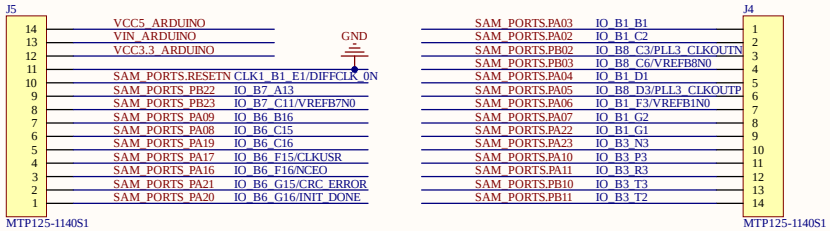


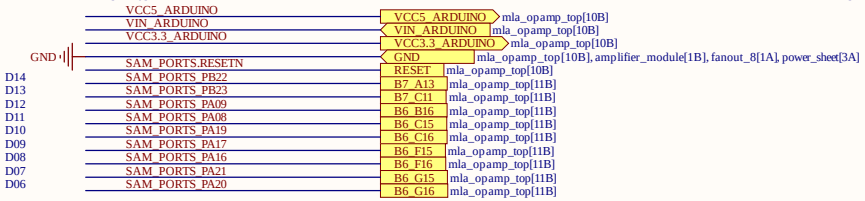
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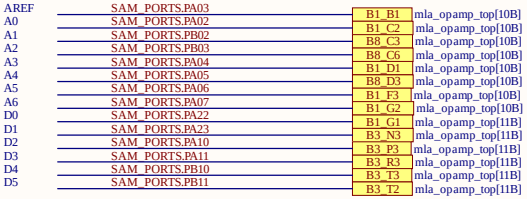
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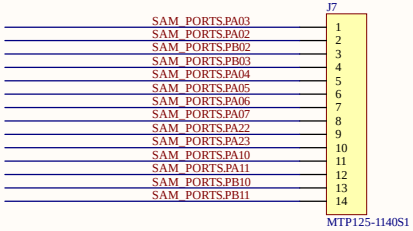

cut bandwidth=10k, delay step1=0us
pz_signal
IC_filter
IC_filter_C1
IC_filter_C2
filter_P_n_of
pz_signal_filtered
R_adding
3224W-2-202E
IC_adding_base
IC_adding_base_C1
IC_adding_base_C2
signal_adding_base
signal_added
Adding_n1
Adding_n2
Adding_n4
Adding_n3
IC_adder
IC_adder_C1
IC_adder_C2
adding_base
signal_adding_base
signal_added
VCC5
GND
Amp_Grp_Mux_P_base
amp_grp1_base_mux
amp_grp2_base_mux
Amp_Grp_Mux
ADG726BSUZ
Amp_Grp_Mux_C1
Amp_Grp_Mux_C2
Amp_Grp_Mux_C3
delay step1=3us
Amp_Grp1_DAC1_C1
Amp_Grp1_DAC2_C1
Amp_Grp2_DAC1_C1
Amp_Grp2_DAC2_C2
Amp_Grp1_DAC1
Amp_Grp1_DAC2
Amp_Grp2_DAC1
Amp_Grp2_DAC2
DAC5568BTER
DAC_SDA
DAC_SCL
PT_ADDR_0
PT_ADDR_1
PT_ADDR_2
PT_ADDR_3
pz_signal_filtered
amp_1_output
amp_grp1_out_filtered
amp_2_output
amp_grp2_out
amp_grp1_base_mux
amp_grp2_base_mux
diff_amp_1_mux_filtered
diff_amp_1_ref
signal_added
Amp_Grp1_P_base_n_of
Amp_Grp2_P_base_n_of
cut bandwidth=95k
Amp_Grp1_filter
Amp_Grp1_filter_R2
Amp_Grp1_filter_C1
Amp_Grp1_filter_C2
Amp_Grp1_filter_C3
delay step1=2us
Amp_Grp1_VR
Amp_Grp1_VR_CCW
Amp_Grp1_VR_W
Amp_Grp1_VR_P
MTP125-114051
3224W-1-104G
Amp_Grp1_amp
Amp_Grp1_amp_C1
LIR3710IMSMTRPBF
Amp_Grp1_amp_P_n_of
Amp_Grp1_ref
Amp_Grp1_ref_reverse
Amp_Grp1_ref_C1
Amp_Grp1_ref_C2
Amp_Grp1_ref_debug
diff_amp_1_ref
delay step2=1us
Amp_Grp2_filter
Amp_Grp2_filter_R2
Amp_Grp2_filter_C1
Amp_Grp2_filter_C2
Amp_Grp2_filter_C3
delay step3=2us
Amp_Grp2_VR
Amp_Grp2_VR_CCW
Amp_Grp2_VR_W
Amp_Grp2_VR_P
Amp_Grp2_amp
Amp_Grp2_amp_C1
Amp_Grp2_amp_n_of
delay step4=1us
Title
Size
A1
Number
Revision
Date
10/11/2026
File
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Sheet of
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800 mil between these two tags

https://docs.arduino.cc/hardware/mkr-vidor-4000/#compatibility

https://docs.arduino.cc/hardware/mkr-connector-carrier/#features

https://docs.arduino.cc/resources/schematics/ABX00022-schematics.pdf

https://docs.arduino.cc/resources/pinouts/ABX00022-full-pinout.pdf

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